Order of operations with whole num[redacted]

1 Calculate:

a $6+4\times 2$

b $6+16\div 4$

c $24\div 6\times 4$

d $22-18\div 6$

e $37-35\div 7+8$

f $36\div 4\times 8+6$

g $18\div 2+7$

h $8\times 5-3\times 2$

i $7+4\times 3-6$

j $11+10\div 5\times 8$

2 Calculate:

a $(7+9)\times 10$

b $7\times(5+4)-50$

c $34-(29+(12\div 3))$

d $(8-2)\times 8$

e $(7-2)\times(6-2)$

f $(10+2)\times(7-2)$

g $[69-(15+24)]\div 3+8\times 6$

h $(11+5)\div(6-2)$

i $[(8+2)\times 6]\div 5$

j $\{[100-(5+15)]+8\}\div 8$

k $[7\times(6-3)]\times[(8+2)\div 5]$

l $[1+(6\times 3-9)]\div[20-(100\div 10)]$

3 Consider the expression $4+7\times 3$.

a State whether the following expressions give the same value as the original expression:

i $(4+7)\times 3$ ii $4+(7\times 3)$ iii $7\times 3+4$ iv $7+4\times 3$

b How do the brackets in $(4+7)\times 3$ change the order of operations from the original expression $4+7\times 3$?

4 Consider the expression $7\div 5+9$.

a State whether the following expressions give the same value is the original expression:

i $7\div(5+9)$ ii $9+7\div 5$ iii $5+9\div 7$ iv $(7\div 5)+9$

b How do the brackets in $7\div(5+9)$ change the order of operations from the original expression $7\div 5+9$?

5 Consider the expression $5-7\times 8$.

a State whether the following expressions give the same value is the original expression:

i $7\times 8-5$ ii $5-(7\times 8)$ iii $(5-7)\times 8$ iv $8\times 5-7$

b How do the brackets in $(5-7)\times 8$ change the order of operations from the original expression $5-7\times 8$?

6 Consider the expression $7 \div 4 - 2$.

a State whether the following expressions give the same value is the original expression:

i $2 - 7 \div 4$ ii $4 - 2 \div 7$ iii $(7 \div 4) - 2$ iv $7 \div (4 - 2)$

b How do the brackets in $7 \div (4 - 2)$ change the order of operations from the original expression $7 \div 4 - 2$?

7 Consider the expression $6 + 19 - 4 \times 3$.

a Evaluate $6 + 19 - 4 \times 3$

b The expressions below are like the expression above, except a pair of brackets has been added. Evaluate the following expressions:

i $6 + (19 - 4) \times 3$ ii $6 + 19 - (4 \times 3)$

iii $(6 + 19 - 4) \times 3$

8 Consider the expression $6 \times 15 - 5 + 3$.

a Evaluate $6 \times 15 - 5 + 3$

b The expressions below are like the expression above, except a pair of brackets has been added. Evaluate the following expressions:

i $6 \times 15 - (5 + 3)$ ii $6 \times (15 - 5) + 3$

iii $(6 \times 15) - 5 + 3$

9 Consider the expression $35 - 6 \times 3 + 2$.

a Evaluate $35 - 6 \times 3 + 2$

b The expressions below are like the expression above, except a pair of brackets has been added. Evaluate the following expressions:

i $35 - (6 \times 3) + 2$ ii $35 - (6 \times 3 + 2)$

iii $(35 - 6) \times 3 + 2$ iv $35 - 6 \times (3 + 2)$